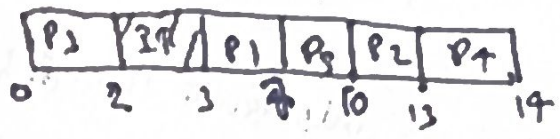
iv) Bounded waiting: A limit must exist on how long a process has to wait before it can enter the critical section.

Problems FCFS (Different Arrival Time)

Type I

Process	AT	BT
P1	3	4
P2	5	3
P3	0	2
P4	5	1
P5	4	3

Gantt chart



Process	AT	BT	CT	WT	TAT
P1	3	4	7	0	4
P2	5	3	13	5	8
P3	0	2	2	0	2
P4	5	1	14	8	9
P5	4	3	10	3	6

WT = TAT − BT

TAT = CT − AT

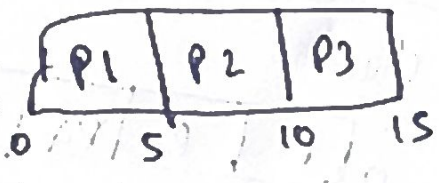
CT → completion time

Avg waiting time = 3.2 ms

Avg TAT = 5.8 ms

Type II (Same Arrival time)

Process	AT	BT
P1	0	5
P2	0	10
P3	0	15



Draw table with same formula

Shortest Job First

Process	AT	BT
A	1	3
P_2	2	4
P_3	1	2
P_4	4	4

Gant chart

	P_3	P_4	P_2	P_1

0 1 3 6 10 14

The A is preemptive.

Non preemptive Consider burst time given

Process	AT	BT
P1	3	1
P2	1	4
P3	4	2
P4	0	6
P5	2	3

P4	P1	P3	P5	P2

0 6 7 9 12 16

Preemptive

Process	AT	BT
P1	0	8
P2	1	4
P3	2	9
P4	3	5

Gant chart

P1	P2	P4	P1	P3

0 1 5 10 17 26

Round Robin

Interval = 2

Same arrival time

Process	AT	BT
P1	0	4
P2	0	5
P3	0	3

Gant chart

P1	P2	P3	P1	P2	P3	P2

0 2 4 6 8 10 11 12

Different AT

Process	AT	BT
P1	0	5
P2	4	2
P3	5	4

P1	P1	P2	P1	P3	P3

0 2 4 6 7 9 11

Priority

Non preemptive

Process	AT	BT	Priority
P1	2	2	2
P2	0	1:0	3
P3	1	3:2	4
P4	3	5	1
P5	4	4	5

P2	P3	P4	P1	P5

0 1 4 9 11 15

Preemptive

Process	AT	BT	Priority
P1	0	11	2
P2	5	28	0
P3	12	2	3
P4	2	10-3	1
P5	9	16	4

Part-II will release on Semester holiday study holidays :-